

ILS Leadership Summit 2026

National Hackathon

Grand Finale: 24-25 October 2026 | Surat

1. Event Overview

The ILS 2026 National Hackathon is a pan-India competition organized as part of the IEEE ILS Leadership Summit 2026. The hackathon is open to student teams from all across India and runs across multiple rounds from July through October 2026, culminating in a Grand Finale at the ILS event in Surat.

The hackathon is designed around two broad tracks. The first is an open track covering rural India challenges, healthcare, and education. The second is a WIE Special Track focused entirely on problems faced by women, including safety, empowerment, and health.

Both hardware and software solutions are welcome. Teams are not restricted to one or the other.

2. Hackathon Tracks and Problem Statements

There are three tracks. Teams must choose one track and one problem statement from within that track. They are not allowed to switch tracks after submission.

Track 1: Bharat Rising (Rural India and EdTech)

This track focuses on real ground-level problems in rural India and education. Teams are expected to build solutions that can work in low-resource environments with limited internet, low-literacy users, or constrained hardware budgets.

Problem 1: The Middleman Killer

In India's agricultural mandi system, farmers consistently earn 20 to 30 percent less than actual market price. This happens because buyers form informal cartels, share price information among themselves, and exploit the farmer's information gap. Build a real-time price discovery and auction platform where farmers can post their produce, see live market rates from multiple mandis, and transact directly. The platform must work in areas with poor connectivity, be resistant to coordinated price manipulation by buyers, and be simple enough for a farmer who has never used a smartphone app to operate with minimal help.

Problem 2: Disaster Response Without Infrastructure

When a flood, earthquake, or cyclone hits rural India, the first thing that fails is connectivity. No internet, no mobile towers, sometimes no power. Relief teams cannot coordinate. Victims cannot send SOS signals. Build a mesh-network emergency communication system using cheap hardware like LoRa or Bluetooth modules that allows people to send location-tagged SOS signals, rescue teams to coordinate movement, and a shared resource map to be built and updated across all connected devices. No internet should be required at any point.

Problem 3: Gamified Vocational Learning for School Children

Children in rural India drop out of school because what they learn inside the classroom has no connection to the world around them. A 10-year-old whose family farms cannot see the point of studying algebra. Build a game-based learning platform with two separate tiers: one for children aged 9 to 12, and one for children aged 12 to 15. Each tier teaches real vocational skills such as basic electrical repair, stitching, small-scale farming techniques, or carpentry through missions, characters, and reward systems.

Track 2: HealthTech

This track focuses on healthcare challenges in India. Solutions should be practical, targeted at real patient populations, and built with constraints like limited doctor availability, low health literacy, and inconsistent access to medical infrastructure in mind.

Problem 4: The Polypharmacy Crisis

An elderly person in India commonly visits three or four different doctors for different conditions. Each doctor prescribes medicines without knowing what the others have prescribed. The result is a cocktail of 8 to 12 medicines that may have dangerous interactions with each other. Build

an AI system that ingests prescriptions from multiple sources, whether scanned, photographed, or typed, detects harmful drug combinations, and then communicates the risk clearly to both the prescribing doctor and the patient in plain language that a 60-year-old with no medical background can understand. The system should also suggest safer alternatives where possible.

Track 3: SheLeads (WIE Special Track)

This track is dedicated to problems faced by women in India. It is the primary track supported by WIE Student Branches across Gujarat Section. All problems in this track are built around real situations that women encounter daily. Solutions must be designed with women as the primary user, not as an afterthought.

Problem 5: Safe Route Navigator

Most navigation apps tell you the fastest route or the shortest route. They do not tell you the safest route for a woman travelling alone at 9pm. Build an app that calculates safety scores for routes based on real crime data, crowd density at different times of day, street lighting, and community-reported incidents. When a woman asks for directions, the app should offer the safest walking or public transport route, not just the quickest one. Route safety scores should update dynamically as new incidents are reported. The app must work in Indian cities where street lighting data or open crime databases may be incomplete.

Problem 6: SheShield Wearable

A basic SOS button that sends a location is not enough. By the time someone checks the message, the situation may already be over. Build a wearable panic device that detects distress passively, using signals like sudden rise in heart rate, abnormal movement patterns, or loud sound events. When distress is detected, it should trigger silently without alerting an attacker, capture audio and video evidence automatically, send live location and a recording to pre-set emergency contacts, alert the nearest police station, and continue operating even if the connected phone is taken away or switched off. The wearable should auto-call the emergency contacts and keep transmitting location and audio until the situation is marked safe.

Problem 7: Skill to Income Platform for Rural Women

Women in rural India have real marketable skills. A woman in a village near Jaipur may make block-printed fabric that a buyer in Bangalore would pay good money for. The gap is entirely about access, not quality. Build a platform where a rural woman can photograph her work, get help listing it with an accurate description and a fair price, find buyers online, coordinate with a logistics partner for delivery, and receive payment in a language and format she understands. The platform should also help her build a digital seller reputation over time so buyers trust her. It must work on a basic smartphone and be fully navigable in at least three Indian languages.

Problem 8: Nutrition Tracker for Pregnant and Lactating Women in Rural India

Standard nutrition apps tell pregnant women to eat spinach salad and Greek yogurt. A pregnant woman in a village in Rajasthan has access to bajra, moth beans, drumstick leaves, and seasonal vegetables. She cannot afford a dietitian. She may not be able to read. Build a diet planner that asks about locally available seasonal produce, the family's weekly food budget, regional cooking habits, and the woman's stage of pregnancy or lactation, and then produces a practical weekly nutrition plan that meets WHO standards for maternal nutrition. All guidance should be available in the local language and presented with visual icons rather than long text, since the user may have limited literacy.

Track 4: Open Industry Problems

Industry to be Contacted

- **Industry will give problem statements they are facing or want solutions**
- **Students will take Problem statement from Industry**

3. Multi-Round Structure

The hackathon runs across four rounds from July to October 2026. Each round eliminates a portion of teams and only the strongest move forward to the next stage.

Round	What Happens	Timeline
Round 1	Registration and Idea Submission: Teams register and submit a Word document covering their chosen problem statement, the problem they are addressing, their proposed solution approach, and the technology they plan to use. No prototype required at this stage.	July 2026
Round 2	PPT Evaluation: Shortlisted teams from Round 1 submit a detailed presentation.	August 2026
Round 3(Grand Finale)	(Online or Regional): Teams that clear Round 2 present a working prototype or demo. This can be a software demo, a hardware prototype, or a combination or Even Design / Simulations. Judging happens online or at designated regional venues. Teams then pitch live either in an online session or in person at the nearest WIE Student Branch location in Gujarat, before a panel of faculty advisors and industry judges. (Judging can be Online/ Fully Offline as per the Condition)	September 2026
Finale(Calling all the Grand Finale Teams)	Finale at ILS 2026: Top teams travel to Surat and present their final solution before a panel of senior IEEE members, faculty, and industry representatives. Winners are announced at the ILS summit.	24-25 October 2026, Surat

Round-Wise Judging Criteria

Criteria	Round 1 and 2	Grand Finale	Finale
Problem Understanding	30%	20%	15%
Solution Creativity	30%	20%	20%
Technical Feasibility	20%	25%	20%
Working Prototype / Demo	N/A	25%	30%
Criteria	Round 1 and 2	Grand Finale	Finale
Real World Impact	20%	10%	15%

4. Team Eligibility and General Rules

Eligibility

- Open to all undergraduate and postgraduate students currently enrolled in any Indian college or university.
- Each team must have a minimum of 2 members and a maximum of 4 members.
- Members can be from different colleges or different disciplines. Cross-institution and cross-discipline teams are encouraged.
- Each student can only be part of one team.
- A faculty advisor is recommended but not mandatory for Round 1.

General Rules

- Teams must build their solution from scratch during the hackathon period. Pre-built products or solutions developed before registration will be disqualified.
- Open-source libraries and publicly available datasets are allowed and encouraged.
- Teams must choose one track and one problem statement. Switching tracks after Round 1 submission is not allowed.
- Plagiarism or copying of any kind will result in immediate disqualification.
- All team communication with organizers must go through the designated team leader.
- The decision of the judging panel is final in all rounds.